Pranjal Arya

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Skills

- **Languages:** C, ARM Assembly, Bash scripting.
- **Protocols:** UART, I2C, SPI, CAN, SD Card, SDIO, eMMC.
- **Processor/SoC :** ARM Cortex - (M4, M33), BeagleBone SoC (ARM Cortex - A8).
- **Miscellaneous:** Git, GDB, OpenOCD, Valgrind, Saleae analyzer, SEGGER SystemView & Ozone, FreeRTOS, TrustZone for Cortex-M, Linux System Programming.

Experience

Member of the embedded platform software team responsible for firmware, BSP and middleware development for PSoC6 and the upcoming family of Infineon PSoC devices.

Infineon Technologies | **Senior Software Engineer** | **April 2023 - Present**

- **Middleware**: Developed Functional Safety middleware library for ARM Cortex-M4 based chips.
- **Bring-up**: Performed pre-silicon and post-silicon firmware bring-up in a bare-metal environment for various ARM Cortex-M4 and M33 based chips. Ensured smooth integration of delivered assets to stakeholders such as boot code, wifi stack, etc.

Infineon Technologies | **Software Engineer** | **July 2020 - March 2023**

- **Peripheral drivers**: Added features in UART, SPI, I2C, CAN, GPIO, IPC, SD Card, SDIO, and eMMC drivers to support ARM Cortex-M4 based chips. Analyzed critical issues reported by customers and internal teams and provided fixes.
- **BSP and CI**: Created scripts and makefiles to automate build and basic sanity testing. Integrated this framework in CI.
- **MbedTLS**: Added PSA driver support and upgraded cryptography hardware acceleration package to support the latest MbedTLS release.

Cypress Semiconductor | **Intern** | **January 2020 - June 2020**

- **Internship project**: Unified peripheral drivers to support all PSoC6 devices and created a common peripheral driver library.

Education

B.Tech. (Electronics & Communication Engineering) - Nirma University, Ahmedabad — CPI: **8.24** | July 2016 - July 2020

HSC — **88.9%** | 2016

SSC — **89.7%** | 2014

Projects

Real-Time Operating System development from scratch — GitHub project link

Implemented FCFS, Round-Robin, Rate Monotonic, Priority, and Sporadic schedulers for Cortex-M based microcontrollers. Also added support to calculate the CPU Utilization of an RTOS. It helped me understand RTOS kernel internals and OS support features of the Cortex-M architecture.

Java-like Garbage collector library for C programs — GitHub project link

Developed a generic C library to detect memory leaks. It has the ability to parse any application's data structures and manipulate them to track the memory blocks malloc'd by the application.

Custom dynamic memory allocation manager in C — GitHub project link

Designed and implemented a custom heap memory manager in C to address the problem of memory fragmentation, see memory usage statistics, and catch memory leaks.

Custom bootloader for STM32L476RG — GitHub project link

Developed custom bootloader for STM32L476RG MCU and host application without using any IDE. Host application in PC interacts with a custom bootloader over UART. The custom bootloader is capable of maintaining information and updating application firmware using an A/B swap mechanism.

SD Card device driver kernel module — GitHub project link

Developed SD Card device driver kernel module for BeagleBone Black (ARM Cortex-A8 based AM335x SoC).

CPU Faults on Arm Cortex-M — GitHub project link

Generated Arm Cortex-M4 system exceptions using software on TM4C123GXL Tiva C series MCU. Wrote a fault handler to dump the faulting stack information over debug prints.

DSP Applications on audio signals using on Arm Cortex-M — GitHub project link

Implemented basic operations such as delay & echo on real-time audio using STM32L476RG MCU. Implemented FIR & IIR filters to remove different frequency components of real-time audio signals using an audio card.

Awards

- **Global Spot Award** Infineon Technologies, January 2023
  Awarded for significant contribution in peripheral driver library development

- **YIP Award** Infineon Technologies, February 2023
  Awarded for an innovative idea under the Infineon idea management program. Implemented a framework which reduced 40% runtime of internal CI.